

Calculus Skeleton Notes

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Average Rates of Change

Goals: Compute average rates of change, slopes of secant lines, and motivate the idea of limits

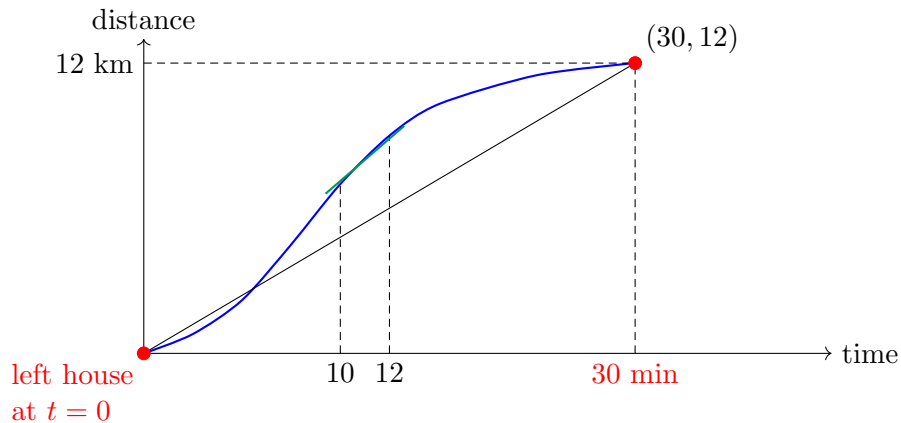
Before Class Videos: None today, starting next class

Example 1: I left my house at time $t = 0$. 30 minutes later, I was at UVic which is 12 km away.

a) What was my average speed?

$$\text{Average Speed} = \frac{\Delta \text{ distance}}{\Delta \text{ time}} = \frac{12 \text{ km}}{0.5 \text{ h}} = 24 \text{ km/h.}$$

b) Sketch a possible graph



c) How could I approximate my speed at $t = 10$ minutes?
Starting at 10 min, I go 1 km in 2 min.

$$\text{Average speed} = \frac{1 \text{ km}}{2/60 \text{ h}} = 30 \text{ km/h.}$$

How can I approximate my speed at 10 min?

Compute average velocity over $[10, 11]$, $[10, 10.1]$, $[10, 10.01]$, etc

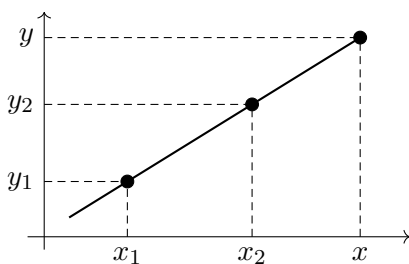
Tell LIDAR story

Future: “Need instantaneous velocity!”

Example 2: What is average rate of change of $f(x) = x^2$ on $[2, 4]$?

$$\text{Average rate of change} = \frac{f(4)-f(2)}{4-2} = \frac{16-4}{4-2} = 6$$

Example 3: Find the equation of the line through (x_1, y_1) and (x_2, y_2) .

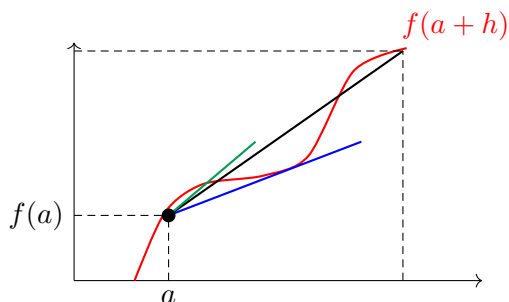


$$\text{Slope} = \frac{y_2 - y_1}{x_2 - x_1} = m, \quad m = \frac{y - y_1}{x - x_1}.$$

$$y - y_1 = m(x - x_1) \implies y = y_1 + m(x - x_1) = mx + \underbrace{[y_1 - mx_1]}_b.$$

Example 4: Consider the graph of a function $f(x)$. How do I write an equation for the secant line through $(x_1, f(x_1))$ and $(x_2, f(x_2))$?

Answer $y = f(a) + \frac{f(b) - f(a)}{b - a}(x - a).$



Slope of secant line

$$= \frac{f(a+h) - f(a)}{(a+h) - a} = \frac{\text{rise}}{\text{run}}.$$

Big Idea: As x_2 gets closer to x_1 the slope of the secant line approaches the slope of the tangent line at x_1 . We need to investigate and define this limiting process more precisely. This DESMOS visualizes the idea: <https://www.desmos.com/calculator/oi7poyhodr>

Suggested after class problems from APEX: None today, but review the Course Outline on Brightspace

1.1 - Limits

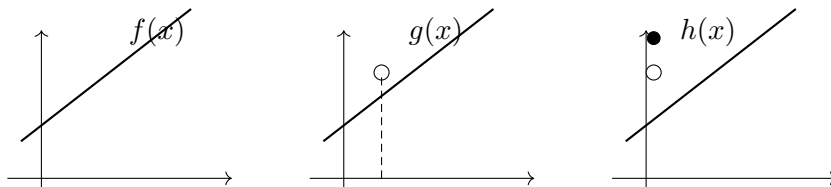
Goals: Describe limits graphically, numerically, and algebraically.

Before Class Videos: 1.1.1 and 1.1.2

Example 1: Consider these three functions:

$$f(x) = x + 1, \quad g(x) = \frac{x^2 - 1}{x - 1} = \frac{(x + 1)(x - 1)}{x - 1}, \quad h(x) = \begin{cases} x + 1, & x \neq 1, \\ 3, & x = 1. \end{cases}$$

How are these similar or different?



Domain: $\mathbb{R}, \quad (-\infty, 1) \cup (1, \infty), \quad \mathbb{R}.$

Same everywhere but $x = 1$, where $f(1) = 2$, $g(1)$ undefined, $h(1) = 3$.

Definition:

$\lim_{x \rightarrow a} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently close to a .

Example 2: Compute the limit as $x \rightarrow 1$ of $f(x)$, $g(x)$, and $h(x)$.

$$\lim_{x \rightarrow 1} f(x) = \lim_{x \rightarrow 1} g(x) = \lim_{x \rightarrow 1} h(x) = 2$$

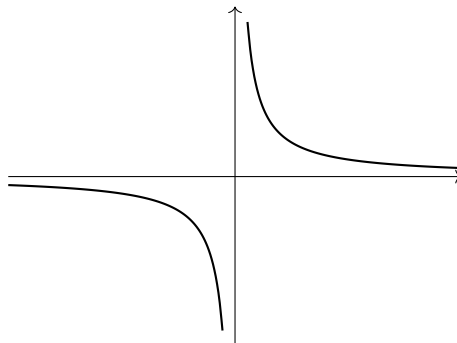
Example 3: Use a table of values to estimate $\lim_{x \rightarrow 1} (x + 1)$.

x	$f(x)$	
0.9	1.9	
0.99	1.99	
0.999	1.999	slope? closer to 2.
1.1	2.1	
1.01	2.01	
1.001	2.001	

Example 4: Use a table of values to estimate $\lim_{x \rightarrow 0} \frac{1}{x}$.

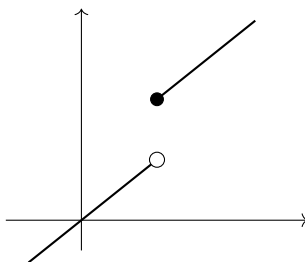
x	$f(x)$	
0.1	10	not getting close to anything!
0.01	100	
0.001	1000	

So limit does not exist!



Example 5: Investigate

$$f(x) = \begin{cases} x, & x < 1, \\ x + 1, & x \geq 1. \end{cases}$$

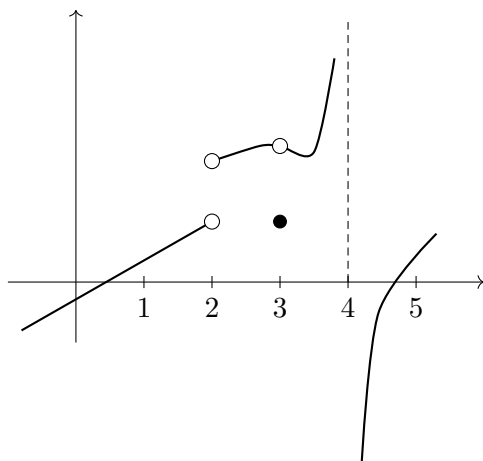


Definition:

$\lim_{x \rightarrow a^+} f(x)$ means $f(x)$ gets arbitrarily close to L for all x sufficiently close to a from the right,
i.e. $x > a$.

$$\lim_{x \rightarrow 1^+} f(x) = \quad \lim_{x \rightarrow 1^-} f(x) =$$

Example 6: Use the graph to evaluate the indicated limits.



$$\lim_{x \rightarrow 2^-} f(x) = \text{---}, \quad \lim_{x \rightarrow 2^+} f(x) = \text{---}, \quad \lim_{x \rightarrow 2} f(x) = \text{---},$$

$$\lim_{x \rightarrow 3^-} f(x) = \text{---}, \quad \lim_{x \rightarrow 3^+} f(x) = \text{---}, \quad \lim_{x \rightarrow 3} f(x) = \text{---},$$

$$\lim_{x \rightarrow 4^-} f(x) = \text{---}, \quad \lim_{x \rightarrow 4^+} f(x) = \text{---}, \quad \lim_{x \rightarrow 4} f(x) = \text{---}.$$

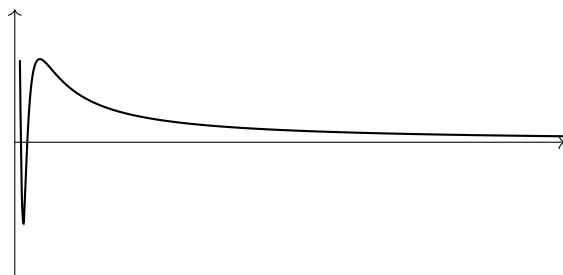
$$\lim_{x \rightarrow 2^-} f(x) = 1, \quad \lim_{x \rightarrow 2^+} f(x) = 2, \quad \lim_{x \rightarrow 2} f(x) \text{ DNE},$$

$$\lim_{x \rightarrow 3^-} f(x) = 2, \quad \lim_{x \rightarrow 3^+} f(x) = 2, \quad \lim_{x \rightarrow 3} f(x) = 2,$$

$$\lim_{x \rightarrow 4^-} f(x) \text{ DNE}, \quad \lim_{x \rightarrow 4^+} f(x) \text{ DNE}, \quad \lim_{x \rightarrow 4} f(x) \text{ DNE}.$$

Example 7: Investigate limits of

$$f(x) = \sin\left(\frac{1}{x}\right)$$



$$\lim_{x \rightarrow 0^+} f(x) \text{ DNE, as } f(x) \text{ gets close to every } y \text{ value in } [-1, 1].$$

Example 8: Graphically investigate limits of $\frac{x^2-1}{x^2-x-2}$

Suggested after class problems from APEX: Section 1.1 problems 9,11,15

1.3 - Finding Limits Analytically

Goals: Evaluate limits using limit laws, algebra, and the Squeeze Theorem.

Before Class Videos: 1.3.1–1.3.3

Definition:

If $f(x)$ gets arbitrarily close to L for all x sufficiently close to a , then

$$\lim_{x \rightarrow a} f(x) = L.$$

For many “nice” functions, evaluating a limit just means plugging in the number. We’ll investigate what functions are “nice” in section 1.5 on Continuity. For example:

Example 1:

- $\lim_{x \rightarrow 2} x^3 =$
- $\lim_{x \rightarrow \pi} \sin(x) =$
- $\lim_{x \rightarrow e} \ln(x) =$
- $\lim_{x \rightarrow 4} \sqrt{x} =$

Theorem:

If L , M , c , and k are real numbers and

$$\lim_{x \rightarrow c} f(x) = L \quad \text{and} \quad \lim_{x \rightarrow c} g(x) = M,$$

then

- | | |
|----------------------------|---|
| 1. Sum Rule: | $\lim_{x \rightarrow c} [f(x) + g(x)] = L + M$ |
| 2. Difference Rule: | $\lim_{x \rightarrow c} [f(x) - g(x)] = L - M$ |
| 3. Constant Multiple Rule: | $\lim_{x \rightarrow c} [kf(x)] = kL$ |
| 4. Product Rule: | $\lim_{x \rightarrow c} [f(x)g(x)] = LM$ |
| 5. Quotient Rule: | $\lim_{x \rightarrow c} \frac{f(x)}{g(x)} = \frac{L}{M}, \quad M \neq 0$ |
| 6. Power Rule: | $\lim_{x \rightarrow c} [f(x)]^n = L^n, \quad n \text{ a positive integer}$ |
| 7. Root Rule: | $\lim_{x \rightarrow c} \sqrt[n]{f(x)} = \sqrt[n]{L} = L^{1/n}, \quad n \text{ a positive integer}$ |

If n is even in the Root Rule, assume that $f(x) \geq 0$ on an interval containing c .

Example 2: Evaluate

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2}.$$

$$\begin{aligned} \lim_{x \rightarrow 2} \frac{x^2 - 2x}{x + 2} &= \frac{\lim_{x \rightarrow 2} (x^2 - 2x)}{\lim_{x \rightarrow 2} (x + 2)} \\ &= \frac{\lim_{x \rightarrow 2} x^2 - \lim_{x \rightarrow 2} 2x}{\lim_{x \rightarrow 2} x + \lim_{x \rightarrow 2} 2} \\ &= \frac{4 - 4}{2 + 2} = 0. \end{aligned}$$

Example 3: Come up with functions f and g such that

$$\lim_{x \rightarrow 3} f(x)g(x) \neq \left[\lim_{x \rightarrow 3} f(x) \right] \left[\lim_{x \rightarrow 3} g(x) \right].$$

The product law requires both limits on the right to exist. For example,

$$f(x) = \frac{1}{x - 3}, \quad g(x) = x - 3.$$

Then

$$\lim_{x \rightarrow 3} f(x)g(x) = 1, \quad \lim_{x \rightarrow 3} f(x) \text{ DNE.}$$

Four Algebraic Tricks

Example 4:

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2}.$$

$$\lim_{x \rightarrow 2} \frac{x^2 - 2x}{x - 2} = \lim_{x \rightarrow 2} \frac{x(x - 2)}{x - 2} = \lim_{x \rightarrow 2} x = 2.$$

The cancellation is valid for $x \neq 2$, which is all that matters when $x \rightarrow 2$.

Example 5:

$$\lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x}.$$

$$\begin{aligned} \lim_{x \rightarrow 0} \frac{(x - 2)^2 - 4}{x} &= \lim_{x \rightarrow 0} \frac{x^2 - 4x + 4 - 4}{x} \\ &= \lim_{x \rightarrow 0} (x - 4) = -4. \end{aligned}$$

Example 6:

$$\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9}.$$

$$\begin{aligned}\lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} &= \lim_{x \rightarrow 9} \frac{\sqrt{x} - 3}{x - 9} \frac{\sqrt{x} + 3}{\sqrt{x} + 3} \\ &= \lim_{x \rightarrow 9} \frac{x - 9}{(x - 9)(\sqrt{x} + 3)} \\ &= \lim_{x \rightarrow 9} \frac{1}{\sqrt{x} + 3} = \frac{1}{6}.\end{aligned}$$

Example 7:

$$\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2}.$$

$$\begin{aligned}\lim_{x \rightarrow 2} \frac{\frac{1}{2} - \frac{1}{x}}{x - 2} &= \lim_{x \rightarrow 2} \frac{\frac{x-2}{2x}}{x - 2} \\ &= \lim_{x \rightarrow 2} \frac{1}{2x} = \frac{1}{4}.\end{aligned}$$

Example 8: Evaluate

$$\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}}.$$

$$\begin{aligned}\lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}} &= \lim_{x \rightarrow 4} \frac{4 - x}{5 - \sqrt{x^2 + 9}} \frac{5 + \sqrt{x^2 + 9}}{5 + \sqrt{x^2 + 9}} \\ &= \lim_{x \rightarrow 4} \frac{(4 - x)(5 + \sqrt{x^2 + 9})}{25 - x^2 - 9} \\ &= \lim_{x \rightarrow 4} \frac{(4 - x)(5 + \sqrt{x^2 + 9})}{(4 - x)(4 + x)} \\ &= \lim_{x \rightarrow 4} \frac{5 + \sqrt{x^2 + 9}}{4 + x} = \frac{5}{4}.\end{aligned}$$

Theorem:

Squeeze Theorem. If $f(x) \leq g(x) \leq h(x)$ for all x in an interval containing a (except possibly at a), and

$$\lim_{x \rightarrow a} f(x) = \lim_{x \rightarrow a} h(x) = L,$$

then

$$\lim_{x \rightarrow a} g(x) = L.$$

Example 9: Evaluate

$$\lim_{x \rightarrow 0} x^2 \sin(1/x^2).$$

Since

$$-1 \leq \sin(1/x^2) \leq 1,$$

we have

$$e^{-1} \leq e^{\sin(1/x^2)} \leq e \implies \frac{x^2}{e} \leq x^2 e^{\sin(1/x^2)} \leq x^2 e.$$

But

$$\lim_{x \rightarrow 0} \frac{x^2}{e} = \lim_{x \rightarrow 0} x^2 e = 0.$$

Therefore, by the Squeeze Theorem,

$$\lim_{x \rightarrow 0} x^2 e^{\sin(1/x^2)} = 0.$$

Example 10: Suppose

$$-x^2 + 4x + 1 \leq g(x) \leq -2x + 10$$

for all x . For what value of a can you determine $\lim_{x \rightarrow a} g(x)$? What is the limit?

The two bounding functions agree when

$$-a^2 + 4a + 1 = -2a + 10,$$

$$a^2 - 6a + 9 = 0,$$

$$(a - 3)^2 = 0,$$

so $a = 3$. Both bounds approach 4, and hence

$$\lim_{x \rightarrow 3} g(x) = 4.$$

1.5 - Continuity

Goals: Determine where a function is continuous, build continuous piecewise functions, and apply the Intermediate Value Theorem.

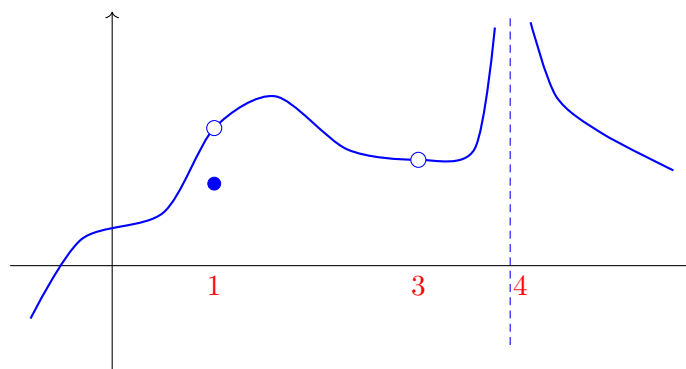
Before Class Videos: Section 1.5 videos 1 and 3

Definition:

A function f is **continuous at** c if all three of the following hold:

1. $f(c)$ exists,
2. $\lim_{x \rightarrow c} f(x)$ exists, and
3. $\lim_{x \rightarrow c} f(x) = f(c)$.

Example 1: Identify the type of discontinuity at each marked x -value.



- There is a jump discontinuity at $x = 1$.
- There is a removable discontinuity at $x = 3$.
- There is an infinite discontinuity at $x = 4$.

Example 2: For what values of a is

$$f(x) = \begin{cases} x + 2, & x \leq 1, \\ ax^2, & x > 1 \end{cases}$$

continuous everywhere?

Example 3: For what values of a and b is

$$f(x) = \begin{cases} -2, & x \leq -1, \\ ax - b, & -1 < x < 1, \\ 3, & x \geq 1 \end{cases}$$

continuous everywhere?

Continuity at the two break points requires

$$\lim_{x \rightarrow -1^-} f(x) = -2 = \lim_{x \rightarrow -1^+} f(x) = -a - b$$

and

$$\lim_{x \rightarrow 1^-} f(x) = a - b = \lim_{x \rightarrow 1^+} f(x) = 3.$$

Hence

$$-2 = -a - b, \quad 3 = a - b.$$

Adding the equations gives $1 = -2b$, so $b = -\frac{1}{2}$. Substitution then gives $a = \frac{5}{2}$.

Combining Continuous Functions

Theorem:

If f and g are continuous at c , then $f + g$, $f - g$, constant multiples of f , and fg are continuous at c . The quotient f/g is also continuous at c provided $g(c) \neq 0$.

Let's justify the sum rule from the limit laws:

$$\lim_{x \rightarrow c} (f(x) + g(x)) = \lim_{x \rightarrow c} f(x) + \lim_{x \rightarrow c} g(x) = f(c) + g(c).$$

Composition of Functions

Example 4: Compute the domain and range of $\sin \sqrt{x}$

Theorem:

Suppose f is continuous at a and g is continuous at $f(a)$. Then the composition $g \circ f$ is continuous at a , and

$$\lim_{x \rightarrow a} g(f(x)) = g(f(a)).$$

Example 5: On what intervals is $\sin\left(\frac{x-2}{x^2+5x+7}\right)$ continuous?

The square-root function is continuous on $[0, \infty)$, and sine is continuous everywhere. Their composition is therefore continuous on $[0, \infty)$.

Example 6: Evaluate

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2}.$$

The numerator and denominator are built from continuous functions, and the denominator is nonzero at $x = 1$. Therefore we may substitute directly:

$$\lim_{x \rightarrow 1} \frac{\sin(\pi x/6)e^{x^3} + x}{x^2 + 2} = \frac{\sin(\pi/6)e + 1}{3} = \frac{e + 2}{6}.$$

Theorem:

Intermediate Value Theorem. If f is continuous on $[a, b]$ and y_0 is between $f(a)$ and $f(b)$, then there is some $c \in [a, b]$ such that

$$f(c) = y_0.$$

Sketch:

Example 7: Sketch examples where the assumptions of the Intermediate Value Theorem are not satisfied.

Example 8: Show that

$$x^3 - 3x - 1 = 0$$

has a solution in $[0, 2]$.

Let $f(x) = x^3 - 3x - 1$. Since f is a polynomial, it is continuous on $[0, 2]$. Also,

$$f(0) = -1 \quad \text{and} \quad f(2) = 8 - 6 - 1 = 1.$$

Because 0 lies between $f(0)$ and $f(2)$, the Intermediate Value Theorem guarantees some $c \in (0, 2)$ for which $f(c) = 0$.

Example 9: Use the intermediate value theorem to show that at some time a baby's height equalled their child's height.

For the equator, compare the temperature at a point with the temperature at its antipode while moving continuously halfway around the equator. The difference reverses sign, so it must equal zero somewhere.

For the parent and child, consider the difference between their heights as a continuous function of time. At the child's birth the parent was taller; earlier in the parent's own life, the parent had the child's birth height. Under the natural comparison of their growth histories, the difference changes sign, so it equals zero at an intermediate time.

1.6 - Limits Involving Infinity

Goals: Investigate the special $\sin(x)/x$ limit, compute limits whose values are infinity, as well as limits at infinity

Before Class Videos: 1.6.1-1.6.3

Example 1: Explore $\lim_{x \rightarrow 0} \frac{\sin x}{x}$.

Example 2: Compute $\lim_{x \rightarrow 0} \frac{\sin(5x)}{\sin(3x)}$.

Example 3: Explore

$$f(x) = \frac{x}{x-2}$$

and find all asymptotes.

Definition:

$\lim_{x \rightarrow a} f(x) = \infty$ if $f(x)$ can be made arbitrarily large for all values of x sufficiently close to a .

Note: A limit being infinite is a particular example of a limit not existing.

Definition:

$\lim_{x \rightarrow \infty} f(x) = L$ if $f(x)$ can be made arbitrarily close to L for all values of x sufficiently large.

Example 4:

$$\lim_{x \rightarrow \infty} \sin(1/x)$$

Example 5:

$$\lim_{x \rightarrow \infty} x \sin(1/x)$$

Example 6: Find all Horizontal Asymptotes for

$$\frac{2x^3 + 3x}{x^4 - x^2}.$$

Example 7: Find all Horizontal Asymptotes for

$$\frac{\sqrt{x^6 + 2x}}{x^3 + x^{1/3}}.$$

Example 8: Find VA's; HA for $e^{1/x}$; graph it!

Example 9: Give example of a function with two HAs, $y = 0$ and $y = 2$, continuous everywhere.
Hint: $\arctan(x)$.

Suggested after class problems from APEX: Section 1.4 #40 and Section 1.6 #10, 19, 27

Derivatives

Goals: Define the Derivative and apply it to initial examples

Before Class Videos: 2.1.1, 2.1.2

Example 1: Recall the velocity problem from the first day of class. If Trefor has some distance function $f(t)$, how fast is he going at 10 minutes? Let's explore graphically and numerically:

Definition:

The derivative of $f(x)$ at c is given by

$$f'(c) = \lim_{h \rightarrow 0} \frac{f(c+h) - f(c)}{h}$$

provided this limit exists. If it does, we say f is differentiable at c .

Example 2: Compute the derivative of $f(x) = x^2$ at $x = 2$.

Definition:

The derivative of $f(x)$ on an open interval I is the function $f'(x)$ given by

$$f'(x) = \lim_{h \rightarrow 0} \frac{f(x+h) - f(x)}{h}$$

provided the limit exists for all x in I . If it does we say $f(x)$ is differentiable on I .

Example 3: $f(x) = 1/x^2$.

Example 4: $f(x) = \sqrt{x}$.

Example 5: Let's explore if the following functions are differentiable at $x = 0$:

a)

$$f(x) = \begin{cases} 2, & x \leq 0, \\ 1, & x > 0. \end{cases}$$

b) $f(x) = |x|$

c) $f(x) = x^{2/3}$

d) $f(x) = x^{1/3}$